

Global Quality Standard GQS304		
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Approved by (printed name and title):
<u>Melissa Seymour</u> Johna Norton – Executive Vice President – Global Quality

Purpose

Distribution is a key interface between Lilly and the customer. Effective distribution controls are necessary to ensure that the right product is sent to the right customer under the right conditions. During distribution, it is critical to ensure handling and transportation methods maintain product quality. The requirements contained in this standard apply to the distribution of finished products, including physician samples and product donations. Distribution processes include the receipt, storage, shipment preparation, and shipping of finished product to the next legal owner (e.g., wholesalers, pharmacies, patients).

Scope

This standard contains requirements related to the management of activities associated with the receipt of finished products from production, other Lilly distribution centers, contract manufacturers, and joint ventures through distribution to affiliates, sales representatives (physician samples), other Lilly distribution centers, wholesalers, consignment pharmacies, and the customer.

This standard pertains to the following types of operations:

- | | | |
|---|----------------------------------|--|
| ☐ materials receipt | ☐ testing | ☒ storage |
| ☐ production | ☐ release | ☒ distribution |
| ☐ packaging and labeling | | |

This standard pertains to the following types of human health products:

- | | |
|--|--|
| ☐ raw materials | ☒ finished products |
| ☐ intermediates | ☒ medical devices |
| ☐ active pharmaceutical ingredients | ☒ drug/device combination products |
| ☐ bulk drug products | ☐ other: N/A |
| ☐ drug products | |

Within this standard, each mention of “product” is inclusive of finished products, medical devices, and drug/device combination products.

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Requirements

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1.0 Distribution Overview

This standard represents the flow of materials from finished product receipt through distribution to customers, affiliates, Lilly distribution centers, wholesalers, private voluntary organizations for donations, and Lilly sales representatives/physicians. The process chart below graphically depicts this process.

The distribution chain is seldom simple and distribution systems can vary enormously. In its simplest form, the chain involves shipment of finished product from the manufacturer to the distribution center to the customer or end user but, in reality, the chain is seldom that short. In its more complex form, the distribution chain may involve a number of storage and transit locations, including airports, docks, and a variety of methods of transport. Therefore, it is important that the distribution chain is adequately designed, managed, and monitored to assure product quality is maintained throughout the process since distribution activities occur after product approval according to GQS102. — Batch Release.

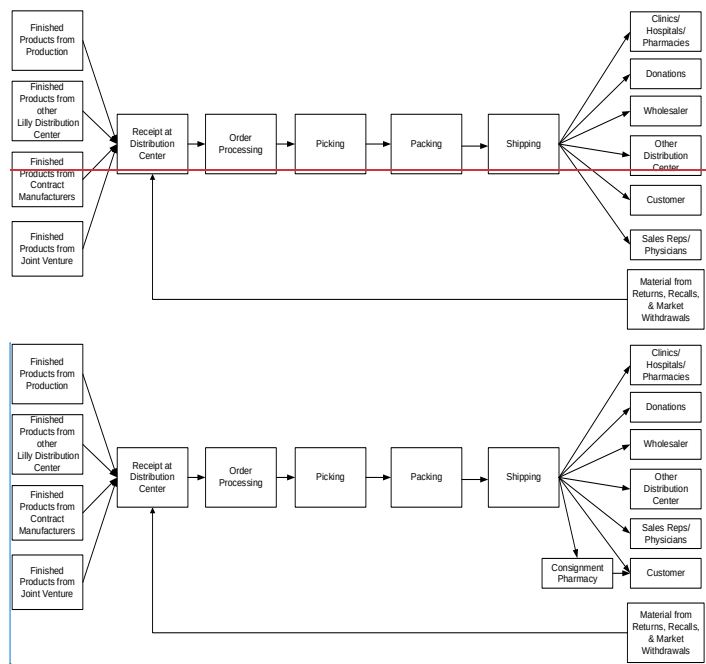


Figure 1: Distribution Product Flow

Commented [CS1]: Updated to include use of consignment pharmacies for distribution of product to customer. See section 6.0 for additional guidance.

Field Code Changed

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1.0 Distribution Overview

1.1 Authorization

- 1.1.1 All Lilly distribution sites must be appropriately licensed/authorized to distribute finished products, as required by local regulations.
- 1.1.2 Finished product must only be sold or distributed to persons or entities that are licensed/authorized to acquire such products according to applicable legislation.
- 1.1.3 A European Union (EU) wholesale distributor must designate a person as Responsible Person (RP). The Responsible Person (RP) must meet the qualifications and all conditions provided by the legislation of the Member State concerned. The Responsible Person (RP) must ensure good distribution practices are in place and are adhered to. In countries outside the EU, the site quality leader for distribution has this responsibility.
- 1.1.4 The quality unit responsible for distribution must ensure that finished products are only distributed to approved markets consistent with the batch release decision (see GQS102, ~~Batch Release~~).
- 1.1.5 Only approved contracted distributors and transport providers, per GQS307, ~~GMP Service Providers and Consultants~~, and appropriately authorized wholesalers can be used. In the United States, a list of wholesale distributors, with the specific products that the distributor is authorized to distribute or a statement that the distributor is authorized for any Lilly product, must be current and available. The list or statement must be approved by the quality unit.

1.2 System

- 1.2.1 There must be a system in place to manage finished product distribution processes. The system must address, at minimum, the following:
 - a. finished product receipt and storage,
 - b. order processing,
 - c. picking and packing processes,
 - d. country-specific distribution designation,
 - e. shipping,
 - f. traceability, documentation, and reconciliation requirements, and
 - g. complaints, recalls, returned product, market withdrawals, imports, exports, and product donations.

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1.0 Distribution Overview

- 1.2.2 -Product must be shipped under appropriate conditions in order to ensure product quality. These conditions include those defined on the storage label, instructions for use and/or those established based on the product's stability data.

Acceptable time periods that a product may be outside of required conditions during routine activities must be defined (e.g., product staging, time required to load and unload product into a vehicle) and must be approved by the quality unit.

Note: Stability Budget documents may be created for refrigerated parenteral products. The Stability Budget document uses product stability data to determine the acceptable Time Out of Refrigeration (TOR). The data is used to plan assembly, packaging, handling, and transportation activities.

- 1.2.3 Sites must have a process for the creation and maintenance of a complete record that traces the ownership of finished product as it is distributed through the supply chain from the manufacturer/importer to the entity (e.g., wholesalers, distributors with a wholesaler license) responsible for selling or supplying the product to the patient or his or her agent (e.g., pharmacies, hospitals, or doctors).

- 1.2.4 Sites must document their process for delivering products requiring special conditions in a risk assessment document. For temperature sensitive products, this includes controls for possible changes in temperature conditions during transport.

- 1.2.5 Sites must have a system in place to segregate products when required. The products must be segregated physically or electronically. Electronic systems used for segregation must be validated and provide security equivalent to physical segregation.

Note: Unless stated otherwise, segregation in this document refers to both physical and electronic segregation.

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1.0 Distribution Overview

- 1.3 Facilities**
- 1.3.1 Distribution facilities must be designed to facilitate the efficient and effective flow of equipment, personnel, and finished product to minimize risks of damage, mix-ups, or substitutions. Additional requirements for facility design can be found in GQS202~~1~~— Facilities.
- 1.3.2 Distribution facilities must be designed, equipped, and operated to meet facility and warehouse security requirements, as well as~~;~~ to minimize the risk of pest infestation per GQS202~~1~~— Facilities.
- 1.3.3 Receiving and shipping bays must be designed to ensure product is protected from weather conditions. The receiving area must be separate from the storage area.
- 1.3.4 Reception areas must be designed and equipped to allow containers of incoming materials to be cleaned where necessary before storage.
- 1.4 General Distribution Requirements**
- 1.4.1 For cold chain distribution, distributors and transport providers specialized in cold chain management of products must be used when available.
- 1.4.2 Quality agreements for management of contracted distributors and transport providers (as determined necessary according to GQS307~~1~~— GMP Service Providers and Consultants) must be in place and must cover any provision for sub-contracting. Quality agreements between Lilly and individual sub-contractors are not required.

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1.0 Distribution Overview

- 1.4.3 Groupage (i.e., consolidation of Lilly products with products from other companies) shipments are only permitted under the following conditions:
- only with non-hazardous and non-perishable goods,
 - when approved in the quality agreement with the distributor,
 - when the temperature and transit time conditions are suitable for all products contained within the shipments (e.g., 2-8°C product must not be grouped with frozen product),
 - when oversight of the transport provider is in place to ensure that the service adheres to the requirements (e.g., periodic monitoring), and
 - when special security substances and/or controlled substances are included in the groupage, these products must not be identifiable as special security substances or controlled substances.
- 1.4.4 The rationale for shipping cold chain and controlled room temperature (CRT) products together under refrigerated conditions (e.g., 2-8°C) must be documented and approved by the quality unit. Shipment of cold chain and CRT products together is not permitted where restricted by the Distribution Temperature Profile (DTP) or local regulations.
- 1.4.5 Finished products must be handled and stored in such a manner to prevent spillage, breakage, contamination and mix ups.

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2.0 Personnel

2.1 Responsibilities of the Responsible Person

- 2.1.1 The Quality function is responsible for ensuring distribution activities are in compliance with applicable requirements. In the European Union (EU), the Responsible Person must carry out their duties in such a way to ensure that the distributor can demonstrate GDP compliance.
- The responsibilities of the Responsible Person include:
- ensuring that a quality management system is in place,
 - focusing on the management of authorized activities and the accuracy and quality of records,
 - ensuring that initial and continuous training programs are implemented and maintained,
 - coordinating and promptly performing any recall operations for medicinal products,
 - ensuring that relevant customer complaints are dealt with effectively,
 - ensuring that suppliers and customers are approved,
 - approving any subcontracted activities which may impact on GDP,
 - ensuring that self-inspections are performed at appropriate regular intervals following a pre-arranged program, and necessary corrective measures are put in place,
 - keeping appropriate records of any delegated duties,
 - deciding on the final disposition of returned, rejected, recalled, or suspect product,
 - approving any returns to saleable stock, and
 - ensuring that any additional requirements imposed on certain products by national law are adhered to.

2.2 Training

- 2.2.1 All personnel involved in distribution must be trained on the requirements of good distribution practices. They must have appropriate competence and experience prior to commencing their tasks per LQS107, Personnel Qualification.
- 2.2.2 Appropriate distribution personnel must be trained on aspects of relevant supply chain controls, intended to avoid suspect product from entering the market.

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2.0 Personnel

- 2.2.3 Personnel must receive specific training if they deal with products that have stringent handling conditions such as hazardous or radioactive products, products that have risks of abuse (e.g., narcotics), and temperature specific products.
- 2.2.4 Personnel involved with the assembly of insulated boxes must be trained on relevant procedures.
- 2.2.5 Personnel must receive initial and continuing training relevant to their role and in accordance with a formal training program.
 The Responsible Person must maintain their competence in GDP through regular training.

2.3 Personnel Hygiene

- 2.3.1 Appropriate procedures relating to personnel hygiene, relevant to the activities being carried out, must be established and observed. Such procedures must cover health, hygiene, and clothing.

3.0 Receipt

3.1 Receipts from Production

Finished product received from production must be segregated and placed in a quarantine status until the product has been approved and released for transfer or shipment to distribution sites.

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3.0 Receipt

3.2 Shipment Receipt

3.2.1 For each incoming shipment, a process for receipt verification must be in place. The quality unit must authorize personnel to perform the examination. Each batch of finished products must be examined to ensure:

- appropriate labeling as to contents,
- the material was shipped correctly (e.g., shipping durations, appropriate qualified container, temperature controlled vehicle),
- agreement between the order, material received, and the delivery note (i.e., supplier's packaging list),
- integrity of the container (e.g., free from damage, no broken seals),
- acceptability of temperature monitoring data (including reconciliation of monitoring devices), if applicable,
- proof of appropriate finished product approval including Qualified Person certification, as required, according to GQS102.
– Batch Release,
- overt safety features on the outer packaging are correct and present, and
- appropriate heat treatment of incoming wooden pallets for pest control was performed and that pallets are free from chemical pesticides and their breakdown products (e.g., Tribromophenols (TBP) and Tribromoanisoles (TBAs)).

NOTE: Heat treatment is the only acceptable method of pest control for wood pallets.

The receipt verification must be documented.

3.2.2 Finished products must be quarantined after receipt, until approval for distribution, using at least one of the following:

- physical segregation,
- procedural controls, (e.g., by use of status labels), or
- a validated computer system (e.g., warehouse or inventory management system).

These processes must be documented.

3.2.3 European sites must segregate finished product received from a non-EU country when it is not intended for the European Market.

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3.0 Receipt

3.3 Approval for Distribution

- 3.3.1 Only the quality unit may approve incoming finished product for distribution. The quality unit must verify the receipt verification documentation. Additionally, the quality unit must examine and verify, at minimum, the following:
- any country-specific status (limited use), and
 - accurate and complete data entry into the validated computer system, if applicable.
- 3.3.2 Where there is potential that the product quality of incoming finished product has been impacted, that product must be quarantined and reported to the quality unit. The issue must be evaluated per ~~GQS104 – Deviation Management~~ [GSOP-104-8, Quality Deviation Management Overview](#). Where relevant, the investigation must include manufacturer notification and review of material trends and history.
- 3.3.3 Per GQS302 – Product Protection and Surveillance, incidents of suspect product must be internally and externally communicated per local regulations.
- Any suspect products received must be physically segregated and stored in a dedicated area away from other finished product. An evaluation per ~~GQS104 – Deviation Management~~ [GSOP-104-8, Quality Deviation Management Overview](#) must be performed.

3.4 Incoming Finished Product Records

- Records must be maintained for the receipt of finished products. These records must include, but are not limited to, the following:
- date and quantity received,
 - purchase order or sales order,
 - name of product,
 - batch or control number and expiration date,
 - description of the product (e.g., dosage form, strength),
 - name and address of the supplier,
 - approval or rejection status,
 - special handling instructions, if applicable, and
 - temperature data, if applicable.

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3.0 Receipt

3.5 Unforeseen Temperature Excursions

If unforeseen temperature excursions outside of established shipping requirement guidelines have occurred, they must be evaluated per [GQS104—Deviation Management](#) [GOSOP-104-8, Quality Deviation Management Overview](#) using the DTP. The responsibility for the shipment and evaluation of associated unforeseen excursions must be described in a quality agreement between sites.

4.0 Storage

Products must be stored under conditions which ensure that optimum quality is maintained throughout their shelf-life, in agreement with their marketing authorization. This section addresses requirements to be met during the storage of the finished product.

4.1 Containers Prior to the storage of products, containers must be cleaned when necessary.

4.2 Storage Areas

- 4.2.1 Finished products stored in drums, bags, or boxes must be stored off the floor (e.g., on pallets). Additionally, these products must be suitably spaced to permit cleaning and inspection. This can also be accomplished by periodic stock rotation (e.g., movement of products on pallets that have not moved in six months).
- 4.2.2 There must be documentation that verifies that Lilly-purchased cardboard pallets or wooden pallets have been treated for pest control. Sites must ensure that wooden pallets have undergone heat treatment for pest control and are free from chemical pesticides and their breakdown products (e.g., Tribromophenols (TBPs) and Tribromoanisoles (TBAs)). Heat treatment is the only acceptable method of pest control for wood pallets. A process must be in place to control the movement of wooden and cardboard pallets.
- 4.2.3 Finished products containing controlled or special security substances must be stored in safe, dedicated, and secure areas according to [LQS303—Controlled Substances](#) [Drugs](#) and Lilly-Designated Special Security Substances. Highly active products (e.g., flammable) must also be stored in safe, dedicated, and secure areas per [HSE-STD-600—Product Stewardship](#). Any applicable international agreements and national legislation (e.g., drug enforcement agency) and associated controls must also be applied.

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4.0 Storage

4.3 Storage Conditions

- 4.3.1 The storage conditions for finished products must be defined in the specification, as required by GQS108, ~~Specifications~~. Finished products must be stored in conditions that meet the specification.
- 4.3.2 Product storage areas must provide adequate protection from extremes of light, heat, and cold as indicated in the product's specification and/or per product registration commitments. Product storage areas must also meet the requirements defined in the official compendium (e.g., USP, JP, and EP, if applicable).
- 4.3.3 Special storage conditions (e.g., operating temperature range, humidity) must be established in accordance with GQS202, ~~Facilities~~, and qualified in accordance with GQS201, ~~Delivery of Physical Assets~~. Temperature mapping must be conducted such that it measures material storage locations most likely to be subjected to temperature extremes (e.g., near overhead lighting, near exterior doors, and near heating and cooling vents).
- 4.3.4 Finished product storage areas must be monitored, and have the appropriate alarm systems for temperature and, where required, humidity control. Storage areas for refrigerated, frozen, and controlled room temperature materials and products must have temperature control. The quality unit must review records of temperature and humidity alarms or excursions, as defined in local procedures for material storage areas.
- All alarm systems and monitoring devices must be maintained according to GQS215, ~~Maintenance~~.

4.4 Segregated Storage Areas

Physical or electronic segregation areas must be provided for the storage of near expiry (per the adjusted expiration date in SAP), expired, or quarantined products.

In addition to the physical segregation requirements in GQS202, ~~Facilities~~, market withdrawn material must be physically segregated. Expired product located in the EU market must also be physically segregated.

If physical segregation areas are used, they must be clearly marked, restricted, and secured on all sides against unintended use.

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5.0 Inventory Control

5.1 Inventory Management

- 5.1.1 All finished products must have an individual inventory record. Each individual inventory record must include the information necessary to determine traceability, batch status, and reconciliation.
- 5.1.2 There must be a defined process of “first expired, first out” (FEFO) for items with expiration dating, unless a customer service agreement requires specific dating periods (e.g., must only ship product with 75% shelf life remaining, WHO Points to Consider for Setting the Remaining Shelf-Life of medical Products Upon Delivery). Changes to FEFO must be temporary, appropriate, documented, and approved by the quality unit.
- 5.1.3 Products beyond their expiration date must be segregated from usable stock and rejected for use.
- 5.1.4 Inventory reconciliation (cycle count) must be performed at least annually. Inventory reconciliation may also be required as part of an investigation of reported inventory discrepancies. Accepted tolerance levels must be defined and set by site operations management and the quality unit. The quality unit must approve adjustment of inventory outside of accepted tolerance. Inventory reconciliation discrepancies outside of accepted tolerance levels must be evaluated per [GQS104—Deviation Management](#) [GSOP-104-8, Quality Deviation Management Overview](#).
Serialized product that cannot be accounted for must be de-aggregated from the parent. The serial number(s) must not be decommissioned. In addition, the record for the serialized product must be updated to identify their current state (e.g., unaccounted for).
- 5.1.5 To facilitate traceability of Lilly products, all distribution sites must have the appropriate processes and procedures in place to ensure that all finished products can be located within 48 hours.

5.2 Quarantine Status

- 5.2.1 A system must be in place to manage finished products remaining in Lilly’s custody where there is potential that product quality has been impacted. This system must call for the immediate quarantine during investigation of the product.

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5.0 Inventory Control

- 5.2.2 To prevent distribution of products that are on quarantine status, the following controls must be in place:
- quarantined products must be segregated,
 - quarantine status must be clearly identified for the product,
 - affected areas and personnel must be advised,
 - product may only be released from quarantine status by the quality unit,
 - product must not be used until the quarantine status has been removed by the quality unit. The authority required to release material from a quarantine (Hold) status must be commensurate with the authority level required for the initial batch release of the product, and
 - a record of the quarantine status must be kept including reference to deviation investigation, if applicable, name of quality unit representative performing the release, and release date.
- 5.2.3 Product released from quarantine may, based on the results of the investigation, be returned to stock, reprocessed, or destroyed. The reprocessing of finished product must follow GQS401~~1~~— Production Practices.

5.3 De-Aggregation of Serial Numbers

If units are picked from a serialized case, the removed units' serial numbers must be de-aggregated from the case.

5.4 Unaccounted for Serialized Product

Serialized product that is unaccounted for per section 5.1.4 must be decommissioned when the time period for a refund for a return has expired. This time period must not be longer than 1 year after expiry unless required by local regulations.

6.0 Establishing Shipping Requirements

The distribution of drug products through an increasingly complex supply chain requires that the distribution supply chain be thoroughly assessed prior to the shipping of finished products. The performance of a distribution risk assessment is a crucial component to ensuring that current and proposed shipping requirements minimize risk to the product during packing and transportation to the customer.

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6.0 Establishing Shipping Requirements

6.1 Distribution Risk Assessment

Shipping practices, specifically the packaging and shipment of temperature sensitive drug products, must be evaluated utilizing a risk based management approach.

To minimize risk to the product, it is each distribution site's responsibility to evaluate the following:

- the individual country regulatory requirements,
- product temperature requirements,
- other hazards (e.g., non-temperature hazards such as chemical, shock, and vibration), as applicable, ~~and~~
- the environment in which the drug product will be transported, ~~and~~
- the complete distribution pathway, from Lilly to the final point of the distribution chain as illustrated in Figure 1. This includes any intermittent stops and/or shipping configuration changes.

This risk assessment must take into account product stability data, methods/modes of transportation, external temperature conditions, duration of shipment, shipment packaging (i.e., qualified container vs. non-protective packaging) and regulatory requirements. The results of the risk assessment will provide the base line for the site's determination of a defined set of shipping requirements for each product or product type.

Commented [CS2]: Added to capture the need to evaluate the use of consignment pharmacies and the change of shipping configuration (pallet to individual unit) that occurs.

6.2 Distribution Temperature Profiles

6.2.1 A Distribution Temperature Profile (DTP) must be developed for each drug product based on stability studies and must define the time and temperature ranges within which a drug product can be transported without adverse effect on its quality. The distribution temperature profile must be created by TS/MS and approved by the quality unit.

NOTE: The Stability Budget document defines acceptable times that a product may be out of refrigeration (TOR) during routine assembly, packaging, and distribution activities in addition to defining the allowance for DTP.

6.2.2 In those instances where a distribution temperature profile is not available (e.g., acquired product), stability data is required in order to determine drug product shipping conditions.

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6.0 Establishing Shipping Requirements

6.3 Shipping Requirements

- 6.3.1 Shipping requirements, derived from the outcome of the distribution risk assessment, must be defined for drug products or drug product families and must include, but are not limited to:
- temperature range and time limits for acceptable shipping conditions that are aligned with the product's stability data or Stability Budget document, if available,
 - acceptable methods /modes of transportation,
 - external temperature conditions,
 - maximum duration of shipment,
 - shipment packaging requirements to ensure product quality,
 - use of qualified containers (e.g., insulated), where required,
 - use of qualified vehicle for product transfer,
 - definition and use of appropriate packing materials (e.g., ice packs, cool packs, gel packs, ~~loadbox~~), ~~and~~
 - adequate protection from external conditions, ~~and~~
 - all shipping configurations are taken into consideration (i.e., switching from pallet to individual unit shipping mid-distribution path).
 - temperature monitoring requirements, where applicable, and
 - local regulatory requirements.

Not all of the above information needs to be located within a single document.

- 6.3.2 When using wooden pallets for shipments, sites must ensure that wooden pallets have undergone heat treatment for pest control and are free from chemical pesticides and their breakdown products (e.g., Tribromophenols (TBPs) and Tribromoanisoles (TBAs)). Heat treatment is the only acceptable method of pest control for wood pallets.

Commented [CS3]: Added to capture the need to evaluate the use of consignment pharmacies and the change of shipping configuration (pallet to individual unit) that occurs.

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6.0 Establishing Shipping Requirements

6.3.3 Methods of transportation and equipment (e.g., air transport containers, chart recorders) must be selected based on the following information:

- the impact of time and temperature conditions on the product's stability (e.g., DTP, Stability Budget document),
- destination,
- route,
- required transport conditions (e.g., temperature),
- climate of the region and seasonal variations,
- delivery schedules, and
- local regulatory requirements.

6.3.4 Temperature sensitive products (e.g., frozen, refrigerated, controlled room temperature) must be shipped in accordance with the defined shipping requirements and local regulatory requirements utilizing:

- temperature-controlled vehicle,
- qualified containers (e.g., insulated containers), when required, or
- monitoring, when the shipping route conditions do not require additional protective measures (i.e., the regional temperature along the shipping route and the duration of the shipment will not impact the quality of the finished product).

Any exceptions must be in compliance with local regulatory requirements, be documented to include rationale, and must be approved by the global quality leader for frozen or refrigerated products or the site/ regional quality leader for CRT products.

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6.0 Establishing Shipping Requirements

- 6.3.5 Additional shipping requirements must be defined to further minimize adverse effects to products being shipped. This includes, but is not limited to, ensuring the following:
- the identity of the product is not lost,
 - product is protected against spillage and breakage,
 - product is not contaminated by unclean vehicles, other product, or rodents, insects, and other pests,
 - product temperature conditions are maintained within acceptable limits, and
 - security measures are in place to prevent unauthorized persons from entering or tampering with vehicles, equipment, or product during shipment and to provide visual evidence of tampering (e.g., trailer seals).
- 6.3.6 Finished products containing special security or controlled substances must be transported in safe and secure containers and vehicles, according to LQS303, — Controlled ~~Substances~~ ~~Drugs~~ and Lilly-Designated Special Security Substances. Products containing highly active material must be transported in safe and secure containers and vehicles in accordance with HSE-STD-600, — Product Stewardship. Any applicable international agreements and national legislation (e.g., drug enforcement agency) and associated controls must also be applied.

7.0 Order Preparation

For many drug products storage and transportation temperatures are a highly significant factor in maintaining the quality of the product throughout the distribution network. Order preparation (e.g., packing requirements) is therefore a key element of control to provide assurance that the product quality is maintained. The requirements contained within this section pertain to the receipt, assignment, and processing of orders for finished product.

7.1 Distribution Orders

- 7.1.1 An order for shipment must be received prior to the distribution of product.

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7.0 Order Preparation

- 7.1.2 Systems must be in place to ensure finished product is assigned to the right order for the right country with the correct labeling. The quality unit must approve these systems (e.g., computer systems, processes, and procedures).
- 7.1.3 Each order must be prepared and shipped in accordance with defined shipping requirements.

7.2 Order Picking Process

There must be a system in place to ensure that the shipment order has been picked correctly. This verification can be done using validated computer systems, or by second person verification. Verification includes, but is not limited to:

- item code,
- batch or control number,
- expiration date,
- storage location,
- quantity,
- description of the product, and
- approval status.

7.3 Order Packing Process

- 7.3.1 Finished product must be distributed in containers (e.g., cardboard shipping cartons, styrofoam containers) that do not have an adverse effect on the quality of the product, and that offer adequate protection from external conditions.
- 7.3.2 The right container and packing material (e.g., ice packs, cool packs, gel packs) must be used according to product temperature requirements, destination, method of transportation, and shipping time. Packaging material must not come into direct contact with the product.
- 7.3.3 There must be a system in place to control the reuse of cool packs to ensure incompletely cooled packs are not used in error. Frozen and chilled ice packs must have adequate physical separation.

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7.0 Order Preparation

7.3.4 SAP temperature conditions and additional instructions, or equivalent, if special transport or storage conditions are required (e.g., keep from freezing, store below 30°C, store at 2 - 8°C), this information must be stated on the shipping label and/or on the outside shipping package, as well as included in the shipping documentation.

7.3.5 Shipping labels applied to outside shipping containers must remain affixed and legible during the shipping of products.

7.3.6 A system to ensure that the shipment order has been packed correctly must be in place. Finished product orders must be reconciled (i.e., order quantity versus actual quantity) prior to shipment. In the event that an order cannot be reconciled prior to shipment, the order must be held and an investigation initiated.

Commented [CS4]: Updated language to be more specific on what fields should be printed on labels. "SAP Temperature Conditions and Additional Instructions or equivalent", instead of Storage Conditions that is now stated. See TR 40647726

8.0 Shipping

The distribution chain may involve, in its most complex form, a number of storage and transit locations, including airports, docks, and a variety of methods of transport, including aircraft. This section addresses requirements to be fulfilled during shipment in order to minimize adverse events on the products being shipped. Monitoring of temperatures in storage facilities and during transportation using calibrated measuring devices is necessary to provide assurance that conditions are under control, and specialized facilities may need to be employed to ensure that the product quality is maintained.

8.1 Shipping and Transportation

- 8.1.1 A record of each shipment must include:
- name of product,
 - a description of the product (e.g., dosage form, strength),
 - the batch number,
 - the name and address of the consignee,
 - the date and quantity shipped, and
 - the transportation company used.

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8.0 Shipping

- 8.2 Monitoring**
- 8.2.1 For shipments of temperature sensitive products that will remain within Lilly control, temperature monitoring must be used during shipping, where required. Temperature monitoring must also be used as directed by the defined shipping requirements and in accordance with local regulatory requirements, where applicable.
- 8.2.2 The shipping site must ensure that all temperature monitoring devices are utilized and maintained according to GQS215, — Maintenance.
- 8.2.3 The locations of product temperature monitoring devices must be recorded in the shipping documentation.
- 8.3 Unforeseen Temperature Excursions During Shipment**
- If unforeseen temperature excursions outside of established shipping requirement guidelines have occurred, they must be evaluated per ~~GQS104—Deviation Management~~ [GSOP-104-8, Quality Deviation Management Overview](#) using the DTP. The responsibility for the shipment and evaluation of associated unforeseen excursions must be described in a quality agreement.

9.0 Returns, Recalls, and Market Withdrawals

- 9.1 General**
- 9.1.1 Returned, impounded, recalled, and market withdrawn finished products must be segregated in accordance with the requirements in section 3.0, — Receipt of this standard.
- 9.1.2 Any suspect products found in the distribution network must be immediately quarantined from other products and must be clearly labeled “not for sale”. Additional requirements for handling these products can be found in GQS302, — Product Protection and Surveillance. Management must be notified of confirmed illegitimate ~~or falsified medicine/~~ product according to ~~LQP-101—GSOP-101-3—Quality~~ Notification to Management (NTM).

Commented [CS5]: Removed: GSOP-101-3 no longer requires NTM for suspect product.

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9.0 Returns, Recalls, and Market Withdrawals

9.2 Returned Product

- 9.2.1 Records must be maintained for each returned product. Documentation must include, but is not limited to:
- product name,
 - name and address of the consignee,
 - batch number and quantity returned,
 - reason for return,
 - disposition of the returned product, and
 - date of final release or rejection.
- 9.2.2 Returned products must be identified as such and held. If the conditions under which returned products have been held, stored, or shipped before or during their return, (or if the condition of the product, its container, carton, or labeling (as a result of storage or shipping)) casts doubt on product quality generally the returned product must be destroyed. If the investigation supports reprocessing (e.g., relabeling, repackaging), then GQS401~~1~~— Production Practices must be followed. If destroyed, the destruction process must be documented and approved by the quality unit.
- 9.2.3 Products that have been returned must be quarantined until a quality decision has been reached regarding disposition in accordance with GQS102~~1~~— Batch Release.

9.3 Recalls and Market Withdrawals

The recall and market withdrawal processes must be managed in accordance with GQS131~~1~~— Recalls and LQS135~~1~~— ~~Product~~~~Market~~ Withdrawal.

9.4 Stolen or Lost Product

- 9.4.1 A ~~notification to management~~deviation must be issued for the suspected theft of product within or outside Lilly in accordance with ~~LQP-101-1— Notification to Management (NTM)~~G SOP-104-8, [Quality Deviation Management Overview](#).
- 9.4.2 Product that has been recovered after loss of control (e.g., stolen or lost shipments) must be quarantined.

Commented [CS6]: No longer used for product theft. Need to update in order to reflect new process.

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9.0 Returns, Recalls, and Market Withdrawals

9.4.3 The impact on quality of the product and the supply chain must be evaluated in accordance with ~~GQS104—Deviation Management~~[GQSOP-104-8, Quality Deviation Management Overview](#).

9.4.4 The global quality leader must approve any proposal to return recovered product to saleable stock. Stolen product that has been recovered cannot be returned to saleable stock for the European market.

9.5 Product Recovery (Salvaging)

Finished products that have been subjected to improper storage conditions (including extremes in temperature, humidity, smoke, fumes, pressure, age, or radiation) due to natural disasters, fires, accidents, or equipment failures must not be used. Whenever there is a question whether finished products have been subjected to such conditions, an evaluation must be conducted per ~~GQS104—Deviation Management~~[GQSOP-104-8, Quality Deviation Management Overview](#).

The global quality leader must approve any decision to use these finished products.

9.6 Returning Product to Saleable Stock

Product returned to saleable stock must be placed such that “first expired, first out” (FEFO) system operates effectively.

9.7 Product Destruction

9.7.1 If finished product is to be destroyed, the destruction process must be documented and approved by the quality unit.

Destruction of pharmaceutical product must be done in accordance with international, national, and local requirements for handling, transport, and disposal.

9.7.2 Serialized product that is destroyed must have their status updated (i.e., decommissioned). The reason for decommissioning must be documented.

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10.0 Import and Export

10.1 Requirements

- 10.1.1 A system must be in place to ensure importation and exportation regulations, as well as appropriate security measures, are met.
- 10.1.2 The authorized quality representative or Qualified Person must approve each batch or partial batch of imported or exported product before release as required by GQS102, ~~Batch Release~~. In the event the batch is exported to another country from its intended final destination the quality representative or Qualified Person that released the product must be informed.
- 10.1.3 If different parts of the batch are released under the same marketing authorization, the quality representative or Qualified Person is responsible for ensuring that records of the importation or exportation of all parts of the batch are kept, and that the distribution of all parts of the batch are traceable.
- 10.1.4 Distributors and importers must co-operate with manufacturers or authorized representatives to ensure traceability of product.

11.0 Product Donations

11.1 General

- 11.1.1 Product donations to private voluntary organizations must be based on an expressed need and consent. Each affiliate must have a documented process to manage product donation requests and must have a designated person responsible for the process.
- 11.1.2 Rules and regulations of each country where donations occur must be followed.
- 11.1.3 The presentation, strength, and formulation of donated drugs should, as much as possible, be similar to those of drugs commonly used in the recipient country.
- 11.1.4 Donated product must have at least six months of expiration dating. Donation of product with less than 6 months expiration dating must have the documented approval of the global quality leader and the managing director (country manager) of the country involved in the donation.

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11.0 Product Donations

11.1.5 Donated product must comply with the country of manufacturer's specification and meet quality standards in both the donor and recipient country.

11.1.6 Product must not be donated unless it is suitable for sale.

12.0 Performance Monitoring and Evaluation

12.1 General

12.1.1 A system must be in place to evaluate and respond to distribution related complaints (e.g., overages, shortages, late shipments, damages during shipment, incorrect deliveries, product theft or loss).

12.1.2 The site quality leader and Global Quality Auditing and Compliance (GQAAC) must be informed of any disruption in the supply of a finished product that could lead to a permanent or temporary interruption in supply of a drug product to the patient according to LQP-112-7, Global Product Shortage and Discontinuation Reporting to Regulatory Authorities.

12.1.3 Distribution management must review information (e.g., reconciliation results, distribution related complaint trends, unforeseen temperature excursions, product theft or loss trends) to monitor distribution performance as part of the management review of quality systems, according to GQS101, Quality Management.

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Revision History

Version Number	Effective Date	List of Major Changes
<u>13</u>	<u>01-Aug-2026</u>	Added references and applicable guidance for the use of consignment pharmacies as part of the distribution chain. Updated guidance on the reporting of product theft. Added guidance on the inclusion of additional instruction(s) on shipping labels. Updated references throughout.
12	01-Apr-2023	Clarified that distribution activities occur downstream of product approval according to GQS102 – Batch Release. Updated references throughout. Removal of non-value-added step in 3.2.4. 5.1.2 Added reference to WHO TRS 1025 Annex 08 Points to Consider for Setting the Remaining Shelf-Life of Medical Products Upon Delivery Apr-2020.
11	01-Jan-2020	Corrected section 3.2.4 to include requirements for receipt at Lilly facilities. Corrected section 5.3 to remove the requirement to decommission in the event that units are removed from a case.
10	01-Jan-2020	Clarified requirements for receipt of serialized product to align with EU distribution agreements. Added temperature data as another piece of information in a product receipt record. Clarified that within this standard, the term product is inclusive of finished products, medical devices and drug/device combination products and added 10.1.4 to address requirements of the EU-Medical Device Regulation (MDR).
9	01-Jul-2019	Added requirements to support the use of a product Stability Budget document. Added descriptions and requirements for unforeseen excursions and acceptable time out of refrigeration. Modified segregation requirement for expired materials located in the EU. Added a review of unforeseen temperature excursions.

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8	01-Jan-2017	Updated to reflect changes to GQS 104. Updated to align with EU GDP requirements language on handling and storage Removed requirement to ensure aggregation on each batch of product Removed Elanco from scope. Updated document format to remove reference citations.
7	01-Apr-2015	Added requirements for receipt, storage and shipping of serialized products. Added requirement for shipping of Special Security Substances. Updated reference from WHO TRS 961 Annex 3 (retired) to WHO TRS 988 Annex 2.
6	01-Sep-2013	Added requirements for new EU GDP: a. handling of recovered stolen product in EU, b. training and hygiene, c. added responsibility for Responsible person and all other personnel, d. change references from EU GMP Part IV Chapter 6 to new EU GDP, and e. cool pack management. Added requirements for drug shortage notification (FDASIA – US). Clarified that standard applies to storage of finished goods at the manufacturing site. Updated references from WHO TRS 908 Annex 4 (retired) to WHO TRS 961 Annex 3.
5	01-Oct-2011	Addition of a requirement specifying that all distribution sites have a process in place to create and maintain records that trace product ownership. Addition of a requirement specifying that all wooden pallets must undergo Heat Treatment for pests and be free of TBPs and TBAs. Clarification between approved service providers and appropriately authorized wholesalers.

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4	01-Nov-2009	Update regulatory references to align with current versions of global regulations. Clarification – requirement to have a list of wholesale distributors is specifically a US requirement. Addition of a requirement specifying that all distribution sites have the capability to locate all finished products within 48 hours. Clarification of shipping requirements for temperature sensitive products. Modifications of sections on shipping and distribution to more accurately reflect the shipping process and improve the flow of the document. Addition of requirements relating to the development of a distribution risk assessment as a necessary component to establishing shipping requirements and practices. Update requirements to include the calibration of temperature monitoring equipment per the applicable standard. Removal of requirement stating imported and exported finished products must be in original packs, as the requirement was removed from the regulation.
3.0	01-Oct-2007	Include the requirement that a list of approved distributors must be available. Include requirement that the quality agreement with the distributor must address allowed goods to be shipped with Lilly product. Correct typographical errors and terminology.
2.0	01-Oct-2005	First version of this global quality standard; see change proposal MLS000204-CHG.
1.0	01-Oct-2005	See change proposal MLS000333-CHG.

Document Approval Signatures

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Reviewer / Approver Additional Details	N/A